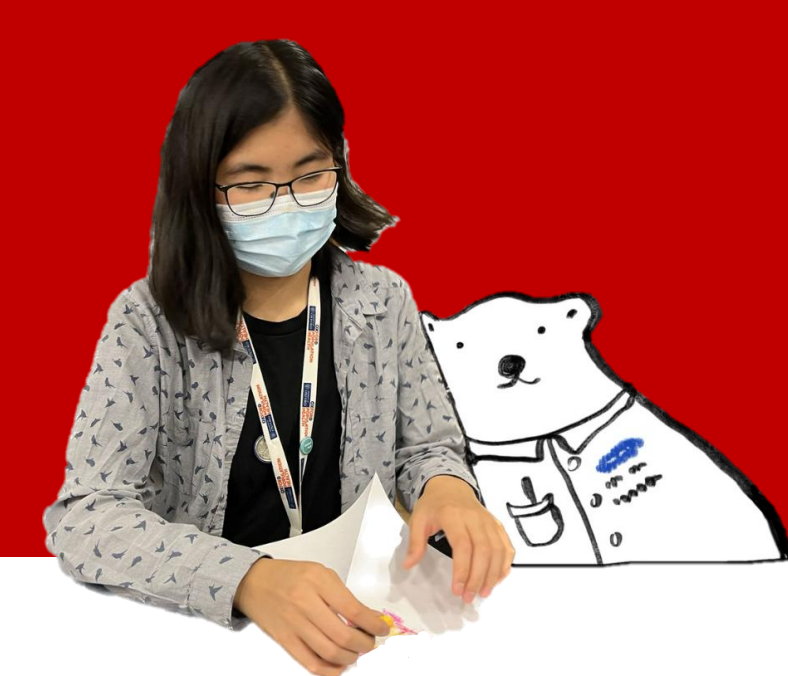


# Collecting stratified volumes from female blood donors could improve donor safety while maintaining donation yield in England



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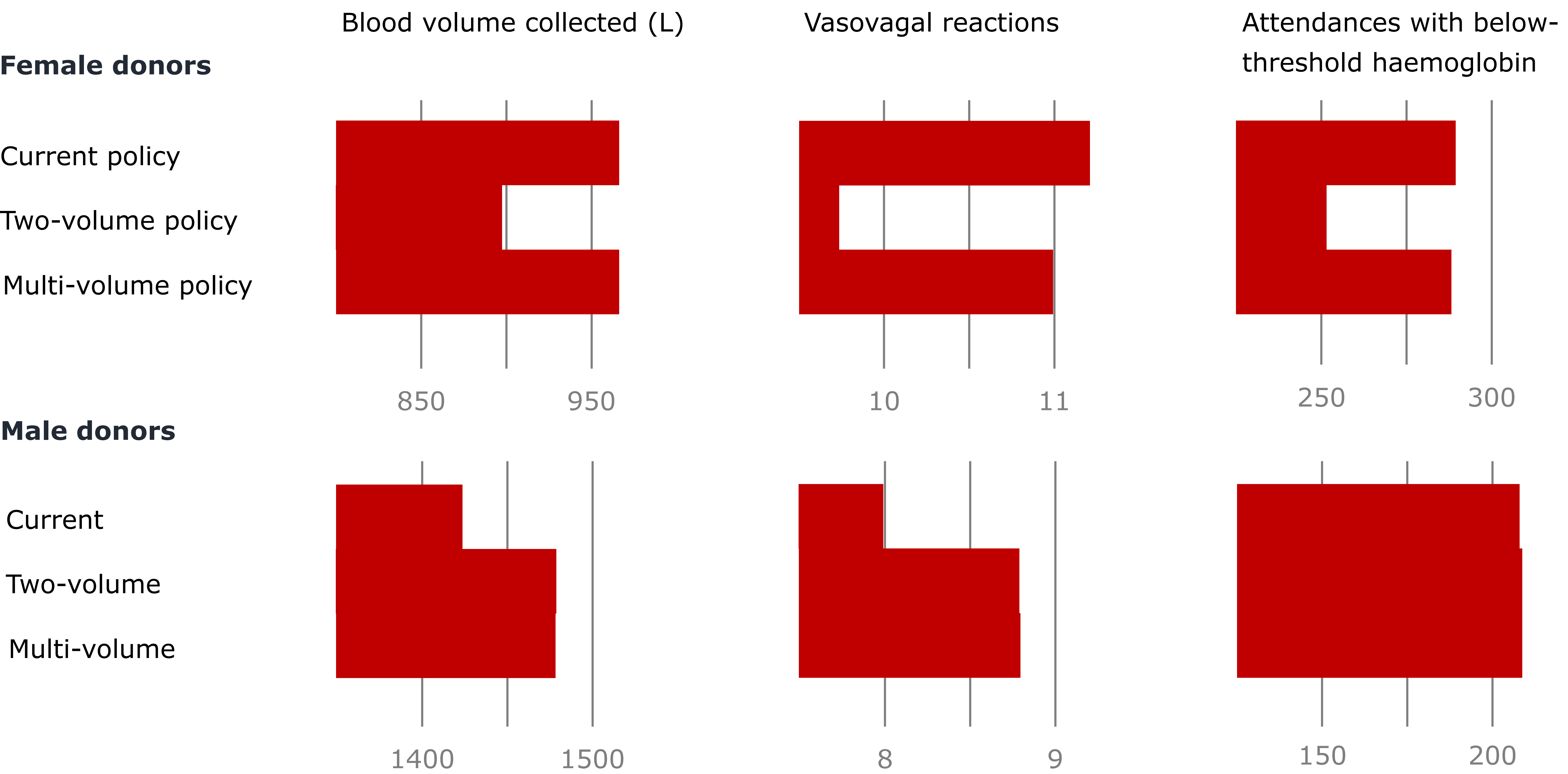
## Background

- Personalised blood collection practices could reduce outcome inequalities among diversifying donor and patient base
- Stratifying blood volume collected by donor estimated blood volume (EBV) could prevent adverse events in smaller (e.g., younger) donors and optimise collection from larger (e.g., Black heritage) donors
- Stratified volumes implemented by multiple global blood services but little evidence available on effectiveness

## Aim

- To evaluate potential impact of stratified volume policies on donor safety and donation yield in England

## Simulation results\*



\*Current policy: universal volume, 475 mL. Two-volume policy: 435 or 495 mL, ensuring loss of  $\leq 13\%$  of donor EBV. Multi-volume policy: 405 to 495 mL in 1-mL increments, ensuring  $\leq 13\%$  EBV loss. All estimates provided per 1,000 donors over 18-month follow-up.

## Interpretation

- Stratified multi-volume collection policy may lead to improved safety and similar yield as current policy among female donors
- Equivalent benefits not observed in male donors, in whom implementation may improve yield but worsen safety
- Stakeholder engagement and pilot testing needed to examine policy feasibility, acceptability, and real-world effectiveness

## Further information

**Preprint:** [yaning-wu.github.io/stratvol-preprint.pdf](https://yaning-wu.github.io/stratvol-preprint.pdf)

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